

AI-Augmented Gamification as a Tool for Interactive Medical Student Education

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Introduction:

The demand for hepatology and transplant hepatology specialists continues to grow, yet early exposure to these fields in undergraduate medical education (UME) remains limited. Specialty choice is strongly influenced by early experiences and mentorship, but pre-clinical students often lack structured opportunities to learn about hepatology, and only 22% of surveyed students reported exposure to a hepatology rotation during clinical phases. This is especially critical as experts predict a 35% shortage of adult transplant hepatologists by 2033, and 62% of transplant hepatology spots went unfilled in 2024 (Kobashigawa et al., 2025).

There is a parallel curricular gap in knowledge in hepatology concepts amongst recently graduated medical students, who report lower confidence in managing chronic liver disease compared with other chronic conditions (Pan et al., 2021). Considering the rising incidence of chronic liver disease, confidence in managing these conditions becomes ever-more important for medical graduates regardless of specialty choice. Based on unpublished data from a survey distributed to medical school curricular deans in 2025, 57.1% of liver transplant center-affiliated medical schools had hepatology preclinical content equivalent in weight to other specialties as compared to 28.6% at non-liver transplant affiliated medical schools. In both medical school groups, <40% reported clinical lectures related to liver transplantation representing a major curricular gap.

Generation Z learners are digital natives, and educational strategies need to be catered to their needs. Unlike prior generations, the traditional lecture platform is less effective for these learners, and methods that include interactivity can be an important strategy to improve learner motivation (Shirazian et al., 2025). Gamification can be an effective strategy for medical education,

especially those with narrative-based designs (Xu et al., 2023). Generation Z learners benefit from individualized feedback (Shirazian et al., 2025), and hence, it is hypothesized that there is a role for AI-integration to augment medical education games to better cater to current and future medical student needs.

Problem Statement:

Despite the impending shortage of hepatologists and the digital-native background of modern medical students, there is a lack of interactive narrative-based tools specifically designed for hepatology education.

Purpose Statement:

The purpose of this study is to compare the changes in self-perceived knowledge of key clinical hepatology topics among US medical students across three medical schools affiliated with liver transplant centers before and after playing an online interactive narrative game.

Research Questions:

Primary:

- How does the implementation of an online interactive narrative game affect self-perceived knowledge of clinical hepatology among medical students across three US medical schools?

Secondary:

1. Is there a statistically significant difference in students' self-reported confidence in managing complications of liver disease before and after playing the game?

2. To what extent do medical students report satisfaction and engagement with narrative-based gamification?
3. Are there differences in self-reported confidence in managing liver disease complications after playing the game between pre-clinical and clinical medical students?

Literature Review

Gamification in Medical Education

Game-based learning has multiple potential benefits including improved learner motivation, utility in formative assessment, and ease of integration into medical education. One example of such an implementation involves quiz-based gamification (Ismail et al., 2019), where in this study medical education gamification can function as learning platforms, guide knowledge gaps, and improve study motivation. Game-based learning does not need to be limited to online-only options as demonstrated by Kinio et al. (2018) whereby a vascular surgery-themed escape room was developed and demonstrated improved learner motivation and learners identifying escape rooms as an asset for curricular inclusion. Narrative based games have also been developed and have been summarized in a systematic review (Xu et al., 2023) which found that games that incorporate storytelling and clinical decision-making elements were particularly engaging formats as it can more closely replicate realistic clinical reasoning. This body of evidence supports interactive narrative-based games as a valid strategy to augment medical education, especially so for medical trainees.

Artificial Intelligence as an Educational Tool

The emergence of large language models has introduced new possibilities for adaptive and personalized learning in health professions education. Gilson et al. (2023) demonstrated that a leading large language model achieved a passing-equivalent score on USMLE Step 1 questions while producing logical clinical justifications across 100% of responses, supporting the potential of these models as interactive medical learning tools. A systematic scoping review by Xu et al. (2024) identified several promising applications of large language models in medical education, including personalized learning plan generation, clinical simulation scenario creation, and formative assessment. Alkhaaldi et al. (2023) surveyed graduating medical students at an international medical school in the United Arab Emirates and found that while formal AI tool use during medical school was minimal, the majority anticipated using these tools during residency training and held broadly optimistic views regarding their role in health care, suggesting an unmet readiness for structured AI integration at the undergraduate level. Russell et al. (2023) proposed a competency framework for AI use by health care professionals, identifying evidence-based evaluation of AI tools as a core training domain, which supports the integration of AI-augmented educational games as a vehicle for both content learning and AI literacy development in a supervised environment.

Identified Gap

The existing literature supports gamified, narrative-based educational tools as effective strategies to improve learner engagement and knowledge acquisition, and highlights the utility of AI-augmented tools in adaptive and individualized medical education. No currently published tool has combined these approaches through narrative gamification with AI-individualized feedback targeting hepatology education for medical students. Given the growing transplant hepatology workforce shortage, demonstrated curricular gaps in hepatology knowledge among graduates, and

the digital-native learning preferences of Generation Z medical students, this represents a meaningful gap in the medical education literature that the present study aims to address.

Methodology

This study utilizes a retrospective pretest-posttest quasi-experimental design to investigate changes in self-perceived knowledge and clinical confidence across medical students after completing an AI-augmented hepatology online game. This study design was selected to avoid revealing the clinical case content prior to students encountering it through gameplay (and hence, no pre-test used for background knowledge at the start of the game). Retrospective pretest-posttest studies (Howard, 1980) have been designed to mitigate the potential for response shift bias that can otherwise be seen in traditional pretest-posttest studies as learners gain an understanding of the assessed criteria during pre-testing that can impact their ability for accurate self-rating post-intervention (Bhanji et al., 2012). Retrospective pretest-posttest study designs have been utilized in medical education (Bhanji et al., 2012) to mitigate this risk of response shift bias.

A text-based AI-augmented browser game has been developed, targeting medical students across three medical schools. The game involves three interactive hepatology clinical cases that students would be expected to encounter during internal medicine clinical rotations. Participants will be recruited using purposive convenience sampling (Creswell & Guetterman, 2025). Inclusion criteria are adult learners currently enrolled in a MD or DO program at one of the three targeted medical schools and are willing to complete the game. Students who have previously participated in game development testing of the game will be excluded. Recruitment will be facilitated through hepatology faculty members identified via the American Society of Transplantation Liver and Intestine Community of Practice (AST LICOP), who will distribute study information and game link

to eligible students via institutional listservs or direct invitation. Institutional Review Board review will be completed prior to study commencement, with a request for a waiver of signed consent due to the study design and impracticality for signed consents.

A sample size of 36 yields 80% power to be able to detect a change of 1.0 in Likert-type scale clinical confidence questions, assuming a standard deviation of 2.0 and an alpha level of 0.05, using a Wilcoxon signed-rank test. With the investigator-derived assumption of a 40% response/completion rate, 90 participants will need to be recruited to achieve this power (30 per site).

The game was developed using Twine through a combination of HTML5 and JavaScript, hosted through GitHub. The AI integration was implemented through two different means: individualized case feedback, and adaptive USMLE-style questions based on an individual learner's performance.

1. AI-Augmented Individualized Case Feedback

The AI engine is designed to compare the student's actions (as identified through a transcript of actions throughout cases) with their user-provided top differential diagnosis per case (free-text reasoning). This allows the system to identify learners who guessed correctly but for the wrong reasons to provide individualized feedback at the end of each case powered by Google Gemini as a virtual attending named Dr. Sato.

2. AI Adaptive USMLE-style Question Generation

As students navigate the game, their critical decisions are tracked in real-time through a transcript backend. If a student misses a key concept (e.g., failing to perform a paracentesis for spontaneous bacterial peritonitis), the system flags that topic for 'Targeted Remediation' at the end of the three cases. Conversely, error-free performance triggers 'General Review.'

During the post-game USMLE-style questioning, the system uses Google's Vertex AI Gemini-Embedding-001 model to embed the identified topic (targeted remediation topic vs general review), queries a Pinecone vector database of investigator provided medical literature (AASLD guidelines, peer-reviewed articles), and retrieves specific source text. This retrieved context is then fed into the large language model to generate high-fidelity, evidence-based board review questions, to minimize hallucination through retrieval-augmented generation. For each of the three clinical cases, the virtual patient's stability, providers reputation based on clinical decision-making, and cost are tracked and integrated into the AI-augmented case feedback. After the end of the USMLE-style questions, learners will complete a post-game survey comprising of Likert-style questions to assess learner's self-perceived confidence in managing each of the clinical cases before and after game completion, game evaluation (including perceived clinical case realism, benefits of immediate individualized AI-generated feedback) and an optional short-answer section on additional feedback regarding the game.

Data will be collected through investigator-developed instruments embedded within the game platform. These have been developed and reviewed by peer hepatologists prior to data collection. Firstly, demographics will be collected to include self-identified gender, medical school location, anticipated graduation year, and intended medical or surgical specialty (if known). Secondly, self-perceived clinical confidence across three hepatology domains (representing each of the three clinical cases) will be assessed using paired retrospective Likert items (consistent with retrospective pretest-posttest design). Thirdly, three Likert questions (each with 5 items per scale) evaluating scenario realism, perceived utility of AI-generated feedback, and comparison against traditional lecture-based learning are included. Finally, an optional free-text field for game feedback is included for qualitative feedback. Survey responses are captured through Google Forms and linked to each participant's in-game session ID (randomly assigned) to integrate survey

data compared to game performance data (patient stability, clinical reputation, cost scores) captured during gameplay. AI-derived adaptive USMLE-style question score result data will also be collected (proportion correct). The paired confidence items (part 1) address the primary research question and secondary research questions one and three. The game evaluation items (part 2) address secondary research question two. The qualitative free-text field and USMLE-style question scores provide further data supporting secondary research questions two and three respectively. The post-game survey containing the questions from parts 1-3 will be provided immediately after learners complete the USMLE-style questions to conclude the game.

Participants' subjective clinical confidence ratings regarding how they felt prior to completing the game will be subtracted from their ratings of how they felt after the game, and these change scores will be compared to a null value of 0 (i.e., no difference) using a Wilcoxon signed-rank test to assess if their confidence significantly changed from before to after. Participants who do not complete the mandatory clinical confidence and scenario realism items (Parts 1 and 2 of the survey) will be excluded from the primary analysis; completion rates across all survey parts will be reported descriptively. As secondary analyses, patient-related data from the clinical cases (e.g., patient stability, reputation, cost, change in clinical confidence) will be compared between groups based on the demographic data collected, such as those who have or have not yet started clinical rotations, as well as USMLE-style question scoring using Wilcoxon rank sum tests. Thematic analysis will be conducted for the qualitative free-text game feedback data (Braun & Clarke, 2006).

Proof of Concept Game Link: <https://hepatology.pages.dev/HepatologyNightShift>

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